

UNIVERSITY OF BONN
CAISA LAB

INTRODUCTION TO NATURAL LANGUAGE PROCESSING

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PROJECT REPORT OF TEAM #23

Modeling Narrative Story Similarity via Representation Learning

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1. Introduction

This project investigates narrative similarity between short stories in natural language processing, focusing on narrative structure rather than surface-level features such as vocabulary, named entities or writing style. We follow the SemEval-2026 Narrative Story Similarity Task¹, which defines narrative similarity along three dimensions: abstract theme, course of action and outcomes.

The task consists of two parts. Task A requires deciding which of two candidate stories is narratively closer to an anchor story. Task B addresses generating story embeddings whose cosine similarity reflects narrative similarity. For Task A, we analyze prompt design effects on large language models; for Task B, we apply contrastive learning to improve baseline embeddings. Overall, we investigate whether task-specific modifications help models capture narrative structure and reduce reliance on surface-level cues.

2. Related Work

Narrative Similarity Prior work in computational narratology ranges from early theoretical frameworks such as Propp’s Morphology of the Folktale [1], which describes narratives in terms of abstract functional structures, to neural approaches that learn character and plot representations from text [2]. More recent hybrid neuro-symbolic systems, such as CoRRPUS [3], combine symbolic reasoning with neural models to support structured story understanding.

Sentence Embeddings Sentence-BERT (SBERT) [4] adapts pretrained BERT models [5] using a siamese architecture to produce sentence embeddings that can be directly compared via cosine similarity. RoBERTa [6] further improved the underlying transformer representations of BERT through optimized pretraining strategies. Since narrative texts and story summaries often exceed typical sentence lengths, document-level representations are commonly obtained by aggregating embeddings of shorter text segments, for example, via mean pooling or weighted averaging.

3. Data Exploration and Preprocessing

This section describes the datasets used, as well as the pre-processing steps that were applied before modelling began.

3.1. Datasets

The shared task provides four datasets (see table 1), which cover different stages of development and evaluation. All of the datasets consist of short narrative summaries, which were either sourced from Wikipedia or written by Large Language Models (LLMs).

In Track A, each data instance consists of an anchor story and two candidate stories and a binary annotation indicates which candidate is narratively more similar to the anchor.

In Track B, stories are provided individually and systems are required to generate a single vector representation for each story.

¹<https://narrative-similarity-task.github.io/>

3.2. Exploratory Data Analysis

We performed a basic descriptive analysis of the narrative texts. As story lengths can vary substantially, ranging from short paragraphs to multi-paragraph summaries, chunk-based processing strategies are required for embedding models.

As seen in figure 3, a frequency analysis of tokens reveals that narrative-driven verbs such as *discover*, *confront*, *escape*, and *return* occur frequently across datasets. This indicates recurring narrative motifs, which may bias models toward superficial similarity if not properly controlled.

The texts also contain many named entities, such as characters, locations and organisations. Although these entities are important from a narrative perspective, they are explicitly excluded from the task’s definition of similarity and thus represent a potential source of noise.

Dataset	Size	Purpose
Sample	39 triples	Debugging and pipeline validation
Development	200 triples	Model evaluation and analysis
Synthetic Training	1900 triples	Training data generated using LLMs
Test	400 triples	Unlabeled; Final evaluation on the leaderboard

Table 1: Overview of the datasets provided for the SemEval task.

3.3. Preprocessing Pipeline

All preprocessing is implemented using the spaCy NLP pipeline, including tokenization, part-of-speech tagging, dependency parsing, and named entity recognition (NER). The same preprocessing workflow is shared across both Tracks for consistency.

We experiment with multiple preprocessing variants to analyze their impact on narrative similarity modeling. These variants are summarized in Table 2.

Variant	Description
Raw (Minimal)	Basic cleaning (e.g., whitespace normalization).
Light Cleaning	Removes non-narrative parenthetical expressions (e.g., actor names).
Entity-based	Uses spaCy NER to replace concrete names with abstract placeholders.

Table 2: Overview of preprocessing variants.

Although entity-based preprocessing is well aligned with the task definition, our experiments have shown that aggressive pseudonymisation can degrade the quality of embeddings due to Named Entity Recognition (NER) errors and the removal of semantically informative tokens. Consequently, the best performance is achieved using raw or lightly cleaned text. Therefore, NER-based preprocessing is retained as an analytical tool rather than as the final embedding input.

4. Methodology

4.1. Track A

Following the task baseline, we used GPT-4o-mini, a pre-trained LLM, accessed via the OpenAI API. We test performance differences through prompt engineering using different prompts. The model is used without any fine tuning.

We designed five prompt variants (PV1-PV5) with different levels of complexity. PV1 (see listing 1) is a short baseline prompt provided by the task. PV2 (see listing 2) introduces explicit definitions and a step-by-step decision procedure. PV3 is a condensed version of PV2 without procedural guidance. PV4 extends PV2 by requiring explicit scoring (0-2) for each narrative dimension, with the final decision having to be based on the aggregated score. PV5 on the other hand is a variation of the baseline prompt, adding a brief instruction to consider narrative dimensions explicitly. These different prompt variants allowed us to systematically study the impact of prompt complexity on narrative similarity judgments.

Lastly, we also investigated the differences of performance by evaluating the model on both the "raw" and the preprocessed (see section 3.3.) narrative text versions. With this we wanted to infer what role named entities play regarding the narrative reasoning of LLMs and how much those influence the performances.

All experiments are implemented in Python using standard scientific libraries for data processing and evaluation (NumPy, Pandas, scikit-learn).

4.2. Track B

Following the task baseline, we used Sentence-BERT with the pretrained `all-mpnet-base-v2` model instead of `all-MiniLM-L6-v2`. We implemented two main approaches: a baseline using pretrained embeddings and a fine-tuned version with supervised contrastive learning.

Baseline

This setup measures how well general purpose sentence embeddings capture narrative similarity without task specific training. Because story lengths vary, each text is split into character based chunks and embedded separately with normalized vectors. This chunk size was chosen to ensure coverage of longer stories while keeping the number of segments manageable. A single story representation is then created using weighted mean pooling over chunk embeddings, giving slightly higher weight to later parts to reflect narrative progression. This weighting scheme is a simple heuristic to emphasize narrative outcomes. In addition, a tail embedding from the final chunks is included to better capture the story outcome, and combined with the full-text representation.

Fine tuning

In the fine tuning step we applied supervised contrastive learning on the baseline model using a triplet loss objective. As illustrated in Figure 4, each training step processes an *anchor* story together with a *positive* story that is narratively closer and a *negative* story that is narratively less similar. All three stories are encoded independently using a shared Sentence-BERT encoder.

Training triplets are constructed directly from the labeled Track A synthetic data. The triplet loss encourages the model to pull anchor and positive embeddings closer together while pushing negative embeddings farther apart in cosine space.

Hyperparameter tuning was performed within a limited scope, involving the variation of learning rates and batch sizes. However, these adjustments only led to minor performance differences.

5. Experimental Setup

For both tracks the performance was primarily measured using Macro F1-Score, as it balances precision and recall and provides a more robust assessment under potential class imbalance.

5.1. Track A

All PVs described in section 4.1. are evaluated in a Zero-Shot and Few-Shot (20/80 split) setting.

Furthermore, to ensure better reproducibility, the model temperature is set to 0 for all tests. Each PV is evaluated over three independent runs on raw data and three independent runs on tokenized data. The performance results were averaged across the three runs and standard deviations were calculated.

No parameter updates or fine tuning was performed on the LLM.

5.2. Track B

Baseline

Each story is split into 350-character chunks (maximum 10), and each chunk is encoded using L2-normalized embeddings. Story representations combine a weighted full embedding, which gives more weight to later chunks, with a tail embedding computed from the last three chunks: $\mathbf{e} = 0.6 \cdot \mathbf{e}_{\text{full}} + 0.4 \cdot \mathbf{e}_{\text{tail}}$.

Predictions use cosine similarity, with tail embeddings as a fallback when the similarity gap is below 0.01. This small margin was introduced as a heuristic threshold to detect uncertain cases where the main embeddings are nearly indistinguishable.

Fine tuning

For fine tuning, the pretrained Sentence-BERT model is further optimized using a triplet loss objective. Anchor–positive–negative triplets are constructed from the labeled Track A synthetic data, where the positive story is narratively closer to the anchor than the negative story.

The model is fine-tuned for a limited number of epochs using cosine similarity as the distance metric. Default optimization settings from the *sentence-transformers* framework are employed, with a low learning rate to reduce the risk of overfitting.

After fine tuning, the resulting embedding model is evaluated using the same inference and similarity computation pipeline as the baseline. This ensures a controlled comparison between pretrained and fine-tuned representations.

6. Evaluation and Results

6.1. Track A

Evaluation Metrics & Data

For the data we used raw and preprocessed versions of the development data for testing this Track.

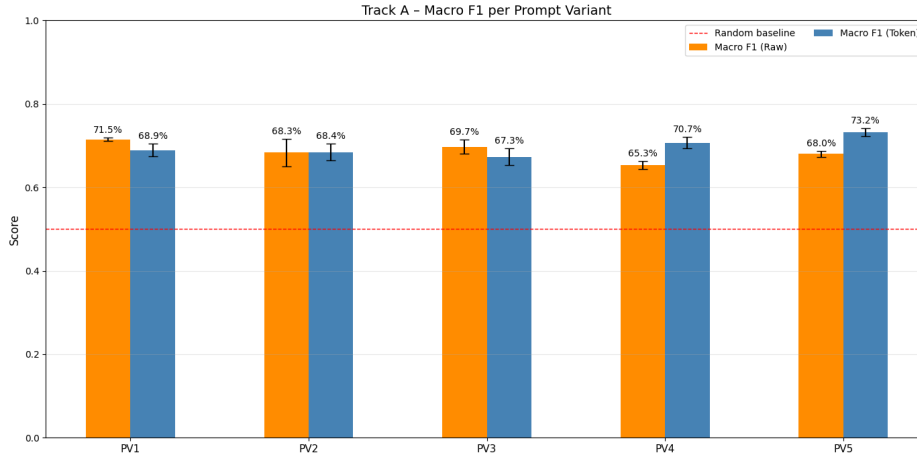


Figure 1: Macro F1-scores for all prompt variants across raw and preprocessed data and their standard deviation.

Zero-Shot Evaluation

As shown in figure 1, on raw data, the baseline prompt PV1 achieves the highest performance with a Macro F1-score of 71.5 % with a low standard deviation of 0.4 %. PV5, with 68 % also has a very low standard deviation of 0.7 %, compared to the other variants. Slightly more detailed prompts such as PV3 perform comparably (69.7 %), while more detailed prompts show reduced performance (PV4) or high standard deviation (PV2).

In contrast, results on preprocessed data show a different trend. Here PV5, the minimally extended baseline prompt with information regarding the narrative similarity dimensions, achieves the best performance overall (73.2 %). More detailed prompts also benefit from preprocessing, outperforming their raw-text counterparts: The standard deviation of PV2 shrinks and the F1-Score of PV4 increases.

Few-Shot Evaluation

On raw data, Few-Shot prompting does not lead to consistent improvements across prompt variants (see figure 2). In most cases the performance changes only slightly compared to the Zero-Shot setting. Standard deviations also increase overall.

An exception is PV4, where a big improvement on raw data (+5.9 %) is seen. This suggests that Few-Shot Learning may be particularly beneficial for prompts that introduce explicit scoring and decision constraints.

For preprocessed data PV2 and PV3 achieve their highest scores in this setting (71.4 % and 70.9 %), closing the performance gap to the raw-data baseline PV1.

6.2. Track B

Baseline

The MPNet-based baseline outperforms MiniLM, achieving a Macro F1 of about 61.5 % on the development set, with tail embeddings providing a small additional gain to around 62.5 %. Despite this improvement, general-purpose embeddings remain limited to surface-level semantics, motivating the use of narrative-supervised fine-tuning.

Fine tuning

Contrary to expectations, fine tuning the embedding model using triplet loss on Track A-derived synthetic data does not lead to improved performance on Track B. Instead, the fine-tuned model performs close to random guessing, achieving a Macro F1 score of 49.5 % on the development set.

To further analyze this behavior, an additional 80/20 split on the development triples was conducted. The fine-tuned model exhibits consistently low performance on both the training and test splits, indicating neither overfitting nor effective learning of the target similarity relations. In contrast, the pretrained baseline remains stable across splits.

These results suggest that the synthetic triplet supervision does not align well with the narrative similarity objective of Track B. Rather than refining the embedding space, the fine tuning process appears to distort relative similarity relationships, leading to degraded performance. Possible explanations include noise in the synthetic labels, objective mismatch between absolute triplet constraints and relative similarity judgments, or partial collapse of the embedding geometry.

Based on these findings, the pretrained MPNet baseline is retained as the final Track B model.

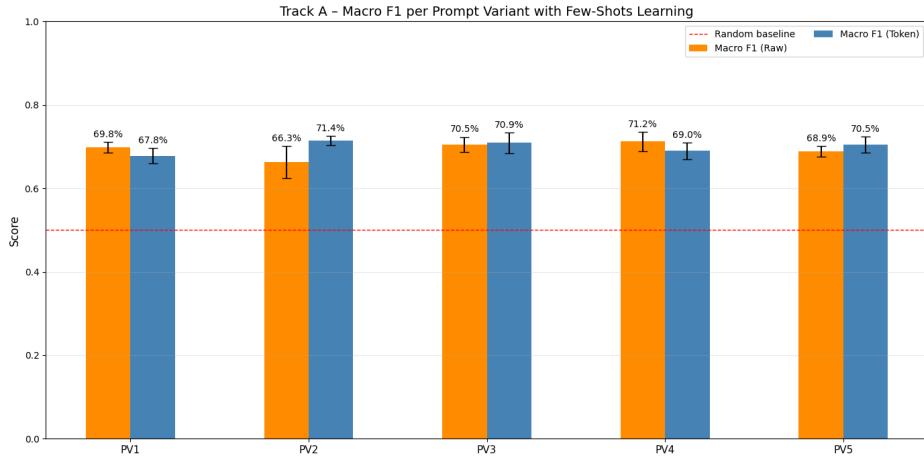


Figure 2: Macro F1-scores for all prompt variants across raw and preprocessed data with additional Few-Shot Learning (20/80) and their standard deviation.

7. Analysis and Discussion

7.1. Track A

The results show that prompt design and data representation moderately affect narrative similarity performance. Simple prompts (PV1, PV5) perform best on raw data, indicating that natural narrative cues such as entities support similarity judgments when prompts are minimally constrained. In contrast, more detailed prompts initially underperform on raw text, suggesting interference with implicit narrative reasoning.

Few-shot learning substantially improves detailed prompts, especially PV4, showing that examples help calibrate explicit decision rules. Tokenization reduces performance for simple prompts but benefits detailed ones by encouraging more abstract narrative comparisons.

Overall, simple prompts are most effective on raw narratives, while detailed prompts benefit from preprocessing and few-shot guidance.

7.2. Track B

The results highlight both the strengths of pretrained sentence embeddings and the challenges of transferring narrative supervision across tasks. The MPNet-based baseline consistently outperforms the MiniLM reference, while chunk-based aggregation and tail embeddings improve stability for longer texts but remain limited to surface-level semantic cues.

Contrary to expectations, narrative-supervised fine tuning using triplet loss does not improve performance on Track B. Across multiple evaluation settings, including an 80/20 split on the development triples, the fine-tuned model performs close to random guessing, indicating a mismatch between the synthetic supervision signal and the embedding-based similarity objective.

This behavior is likely caused by an objective mismatch between triplet loss and the fine-grained relative similarity judgments required in Track B. While triplet loss enforces absolute distance constraints, noisy or coarse synthetic labels may distort the embedding geometry rather than refine it.

Overall, these findings emphasize the robustness of pretrained sentence encoders as strong baselines and suggest that narrative-supervised fine tuning may degrade performance when training objectives are not closely aligned with evaluation criteria.

8. Conclusion

This work studied narrative story similarity in the SemEval-2026 shared task, covering both comparative judgment (Track A) and representation learning (Track B).

For Track A, simple baseline prompts already achieve strong performance on raw data, indicating that LLMs can leverage narrative cues without complex procedural guidance. While structured prompts and Few-Shot Learning improve results in specific cases, overall gains remain limited, suggesting that prompt engineering alone provides only modest benefits.

For Track B, pretrained sentence embeddings prove robust, with MPNet consistently outperforming MiniLM. Fine-tuning Sentence-BERT with synthetic triplet supervision degrades performance, highlighting a mismatch between the fine-tuning objective and narrative similarity judgments.

Overall, task-aware interventions can influence narrative similarity modeling, but their effectiveness depends strongly on the modeling paradigm and supervision strategy. Pretrained models remain strong baselines, while naive supervised fine-tuning may be ineffective or harmful.

9. Future Work

Future work could explore more robust ways of enforcing narrative reasoning in LLMs, for example through lightweight instruction-tuning or fine-tuning on narrative similarity annotations. Such approaches may reduce reliance on surface-level cues observed in baseline prompting, while avoiding the performance degradation associated with over-prompting, as described in the Few-Shot Dilemma [7].

Beyond prompting, future research could investigate dimension-specific narrative representations (theme, action, outcome) or event-centric embeddings to better align with annotation guidelines. Finally, extending the evaluation with human judgments or dimension-wise metrics would provide deeper insight into model behavior and narrative understanding.

10. Task Allocation between Members

Member	Tasks
Moritz Gilges	Preprocessing & Track B: Fine Tuning
Juri Hayashi	Track B: Baseline & Group coordination
Seyed-Arsam Islami	Track A
Mohamed Abouelella	Research: "Related Work" (Because of a sudden hospitalization and a surgery)

Table 3: The task allocation between the team members for this project.

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- [2] David Bamman, Brendan O’Connor, and Noah A Smith. Learning latent personas of film characters. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics*, pages 352–361, 2013.
- [3] Yizhou Dong, Lara J Martin, and Chris Callison-Burch. Corrpus: Code-based structured prompting for neurosymbolic story understanding. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13173–13193, 2023.
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- [5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.
- [6] Yinhan Liu et al. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- [7] Yongjian Tang, Doruk Tuncel, Christian Koerner, and Thomas Runkler. The few-shot dilemma: Over-prompting large language models, 2025.

11. Appendix

11.1. Data analysis - Wordcloud



Figure 3: A wordcloud representing the most common words in the given datasets.

11.2. Prompt variant 1

```
# PV1: Baseline prompt
system_prompt = "You are an expert on stories and narratives. Tell us  
which of two stories is narratively similar to the anchor story."  
user_prompt = f"{few_shots_examples}Anchor story: {anchor}\n\nStory A: {  
text_a}\n\nStory B: {text_b}"
```

Listing 1: The short baseline prompt (PV1) given by the task.

11.3. Prompt variant 2

```
# PV2: Well defined task and procedure
system_prompt = "You are an expert annotator for narrative similarity.
    You strictly follow the official Narrative Similarity Annotation
    Guidelines."
user_prompt = f"""Your task is to decide which of two stories is
    narratively more similar to the Anchor story.

Narrative similarity is defined ONLY by:
1. Abstract Theme: the underlying central ideas, problems, or motifs.
2. Course of Action: the ordered sequence of key events and conflicts.
3. Outcomes: the final resolution, character fates, or moral lesson.

Do NOT consider:
- writing style
- lexical overlap
- named entities
- time period or location

Procedure:
1. For each story, identify:
    a) Abstract Theme
    b) Course of Action (as an ordered list of key events)
    c) Outcome
2. Compare Story A and Story B to the Anchor along each dimension.
3. Decide which story is overall more narratively similar to the Anchor.

{few_shots_examples}

Anchor story:
{anchor}

Story A:
{text_a}

Story B:
{text_b}
"""
```

Listing 2: Extract of the used code for the prompt variant 2 which defines the task and the procedure detailed.

11.4. Finetuning workflow Track B

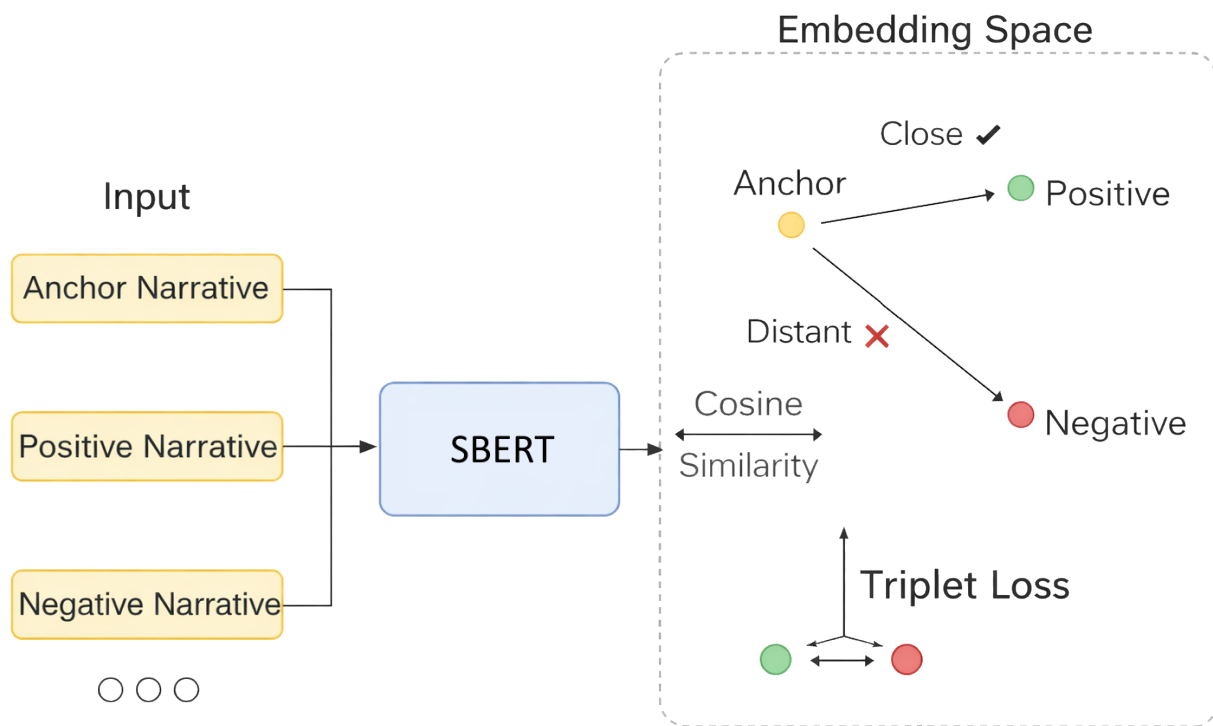


Figure 4: Workflow used to fine-tune Track B using contrastive learning.